NUMBER SYSTEM

Complete Concept + Formula + Example Notes

Beginner to MCA / DSA / Programming level

This PDF is designed as a structured revision guide. Every major topic is defined first, then explained with examples, formulas, properties, and programming connections.

Level	Topics
Foundation	Types of numbers, number line, factors, multiples, primes, divisibility, HCF/GCD, LCM
Arithmetic	Fractions, decimals, rational/irrational numbers, powers, roots, absolute value, remainders
Computer Science	Binary, octal, decimal, hexadecimal, conversions, binary arithmetic, complements, bitwise operations
DSA / Number Theory	Euclidean algorithm, sieve, fast exponentiation, modular arithmetic, modular inverse, Euler phi, Fermat, CRT

How to Use These Notes

If you are rebuilding your mathematics foundation, do not try to memorize formulas first. Follow this sequence: understand the definition -> work through a small example -> learn the property/formula -> solve 5-10 questions without looking at the answer.

- Phase 1: Sections 1-10. Build the mathematical vocabulary.
- Phase 2: Sections 11-17. Learn powers, roots, divisibility, remainders and patterns.
- Phase 3: Sections 18-26. Learn number systems and binary concepts for MCA/computer science.
- Phase 4: Sections 27-35. Study number theory when you are comfortable with DSA.

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1. What is a Number?

Definition: A number is a mathematical object used to count, measure, label, compare, or represent quantities. Examples include 0, 7, -3, 1/2, sqrt(2), and pi.

Digits vs Numbers

A digit is a single symbol used to write numbers. In the decimal system the digits are 0 through 9. A number may contain one or more digits. For example, 7 is both a digit and a number, while 582 is a number made from the digits 5, 8 and 2.

Place Value

In base 10, the value of a digit depends on its position. For 582: 5 is in the hundreds place, 8 is in the tens place, and 2 is in the ones place.

```
582 = 5 x 10^2 + 8 x 10^1 + 2 x 10^0
```

Remember: Always distinguish a digit from a number. This becomes important when learning binary and hexadecimal systems.

2. Types of Numbers

Natural Numbers (N)

Definition: Natural numbers are the positive counting numbers: 1, 2, 3, 4, ... Some textbooks include 0 in natural numbers, so check the convention being used.

Whole Numbers (W)

Definition: Whole numbers are the non-negative integers: 0, 1, 2, 3, ...

Integers (Z)

Definition: Integers include negative whole numbers, zero, and positive whole numbers.

> ..., -3, -2, -1, 0, 1, 2, 3, ...

Rational Numbers (Q)

Definition: A rational number can be written as p/q where p and q are integers and q is not zero.

> 1/2, -7, 0.25, 3/4

Irrational Numbers

Definition: An irrational number cannot be expressed as p/q using integers p and q. Its decimal expansion is non-terminating and non-repeating.

> sqrt(2), sqrt(3), pi

Real Numbers (R)

Definition: Real numbers include every rational and irrational number. They can be represented on the real number line.

Complex Numbers

Definition: A complex number has the form a + bi, where a and b are real numbers and i^2 = -1. Complex numbers are beyond the core Number System revision but are useful later in mathematics.

Set	Meaning	Examples
N	Natural	1, 2, 3...
W	Whole	0, 1, 2...
Z	Integers	-2, -1, 0, 1, 2...
Q	Rational	1/2, -3, 0.75
Irrational	Not p/q	sqrt(2), pi
R	Real	All rational + irrational

> N is contained in W, W is contained in Z, Z is contained in Q, and Q is contained in R. Irrational numbers are also contained in R.

3. Number Properties and Special Numbers

Even Numbers

Definition: An integer divisible by 2 is even.

> n = 2k, where k is an integer

Odd Numbers

Definition: An integer not divisible by 2 is odd.

> n = 2k + 1

Prime Numbers

Definition: A prime number is a positive integer greater than 1 with exactly two positive factors: 1 and itself.

Examples: 2, 3, 5, 7, 11, 13. The number 2 is the only even prime.

Composite Numbers

Definition: A positive integer greater than 1 that has more than two positive factors is composite.

Co-prime Numbers

Definition: Two numbers are co-prime if their GCD is 1. They do not have to be prime themselves.

> 8 and 15 are co-prime because GCD(8,15) = 1.

Consecutive Numbers

Definition: Numbers that follow each other without gaps. Examples: 4,5,6 or -2,-1,0,1.

Perfect Number

Definition: A positive integer equal to the sum of its proper positive divisors. Example: 6 = 1 + 2 + 3.

Perfect Square and Perfect Cube

A perfect square is an integer that can be written as n^2. A perfect cube can be written as n^3.

> 25 = 5^2; 64 = 4^3

4. Factors and Multiples

Factor

Definition: A factor of a number divides it exactly with remainder 0.

Example: Factors of 12

1 x 12 = 12

2 x 6 = 12

3 x 4 = 12

Therefore the positive factors are 1, 2, 3, 4, 6, 12.

Multiple

Definition: A multiple of a number is obtained by multiplying it by an integer.

> Multiples of 5 = 5x1, 5x2, 5x3, ... = 5,10,15,20,...

Prime Factorization

Definition: Prime factorization expresses a positive integer as a product of prime numbers.

> 60 = 2 x 2 x 3 x 5 = 2^2 x 3 x 5

Remember: Factors are finite for a positive integer; multiples continue indefinitely.

5. Divisibility Rules

Definition: A divisibility rule is a quick test for deciding whether one integer divides another without performing the full division.

Divisor	Rule	Example
2	Last digit is 0,2,4,6,8	124 is divisible by 2
3	Sum of digits is divisible by 3	123 -> 1+2+3=6
4	Last two digits divisible by 4	1316 -> 16
5	Last digit is 0 or 5	275
6	Divisible by both 2 and 3	114
8	Last three digits divisible by 8	1512 -> 512
9	Sum of digits divisible by 9	729 -> 18
10	Last digit is 0	430
11	Difference of alternating digit sums is divisible by 11	121 -> (1+1)-2=0
12	Divisible by both 3 and 4	144
25	Last two digits are 00,25,50,75	875
100	Last two digits are 00	1200
125	Last three digits divisible by 125	1875 -> 875

Remember: Divisibility rules are especially useful for HCF/LCM questions, prime testing, and programming problems involving factors.

6. Prime Numbers and Prime Factorization

Definition: A prime number has exactly two positive divisors. A composite number has more than two.

Testing if a Number is Prime

To test whether n is prime, you only need to test possible divisors up to sqrt(n). If no integer from 2 through floor(sqrt(n)) divides n, then n is prime.

Example: Is 29 prime?

sqrt(29) is about 5.38, so test 2, 3 and 5.

29 is not divisible by 2, 3 or 5.

Therefore 29 is prime.

Fundamental Theorem of Arithmetic

Statement: Every integer greater than 1 can be represented uniquely, apart from the order of factors, as a product of primes.

```
84 = 2^2 x 3 x 7
```

Twin Primes

Definition: A pair of primes that differ by 2. Examples: (3,5), (5,7), (11,13).

7. HCF / GCD

Definition: HCF (Highest Common Factor), also called GCD (Greatest Common Divisor), is the largest positive integer that divides all the given numbers.

Method 1: Factor Listing

List factors of each number and select the greatest common factor.

Method 2: Prime Factorization

```
36 = 2^2 x 3^2; 48 = 2^4 x 3
```

For the GCD, take common primes with the smallest powers.

```
GCD(36,48) = 2^2 x 3 = 12
```

Method 3: Euclidean Algorithm

Repeatedly replace (a,b) with (b, a mod b) until the remainder becomes zero. The last non-zero remainder is the GCD.

```
GCD(a,b) = GCD(b, a mod b)
```

GCD(48,18)

48 mod 18 = 12

18 mod 12 = 6

12 mod 6 = 0

Therefore GCD = 6.

8. LCM

Definition: The Least Common Multiple is the smallest positive integer that is a multiple of all given numbers.

Prime Factorization Method

> 12 = 2^2 x 3; 18 = 2 x 3^2

For LCM, take every prime appearing with its largest power.

> LCM(12,18) = 2^2 x 3^2 = 36

Important Relationship

> For positive a and b: GCD(a,b) x LCM(a,b) = a x b

When LCM is Useful

- Finding when repeating events occur together.
- Solving fraction addition/subtraction.
- Finding a common multiple in scheduling problems.
- Programming problems involving cycles.

9. Fractions

Definition: A fraction represents a part of a whole and is written p/q, where q is not zero. p is the numerator and q is the denominator.

Types

- Proper fraction: numerator has smaller magnitude than denominator, e.g. 3/5.
- Improper fraction: numerator has magnitude greater than or equal to denominator, e.g. 7/5.
- Mixed number: whole number plus a proper fraction, e.g. 1 2/5.
- Equivalent fractions: different-looking fractions with the same value, e.g. 1/2 = 2/4 = 4/8.

Operations

a/b + c/d = (ad + bc)/(bd)

a/b - c/d = (ad - bc)/(bd)

(a/b)(c/d) = ac/bd

(a/b) / (c/d) = ad/bc, where c is not zero

Example: 2/3 + 1/6

LCM of 3 and 6 is 6.

2/3 = 4/6.

4/6 + 1/6 = 5/6.

10. Decimals

Definition: A decimal represents a number using a base-10 place-value system.

Terminating Decimal

A decimal that ends after a finite number of digits. Example: 0.25.

Non-terminating Recurring Decimal

A decimal that continues forever but repeats a pattern. Example: 1/3 = 0.3333...

Non-terminating Non-recurring Decimal

A decimal that continues forever without a repeating pattern. Such decimals are irrational when they represent real numbers such as sqrt(2).

Fraction to Decimal

> 1/4 = 0.25; 3/8 = 0.375

A rational number has a terminating decimal exactly when, after simplification, its denominator contains no prime factors other than 2 and 5.

> 1/8 terminates because 8 = 2^3. 1/3 does not terminate because 3 is not a factor of 10.

11. Rational and Irrational Numbers

Rational Number

Definition: A real number that can be written as p/q, with p and q integers and q != 0.

Irrational Number

Definition: A real number that cannot be represented as p/q. Its decimal representation is non-terminating and non-repeating.

Useful Facts

- The sum of two rational numbers is rational.
- The product of two rational numbers is rational.
- Rational + irrational is irrational.
- Non-zero rational x irrational is irrational.
- Two irrational numbers can have a rational sum or product.

Important example

sqrt(2) is irrational.

-sqrt(2) is irrational.

sqrt(2) + (-sqrt(2)) = 0, which is rational.

12. Real Numbers and Number Line

Definition: The real numbers are all numbers that can be placed on the real number line.

The number line orders numbers from smaller to larger as we move from left to right.

```
... < -3 < -2 < -1 < 0 < 1 < 2 < 3 < ...
```

Comparing Negative Numbers

For negative numbers, the number closer to zero is greater. Therefore -2 > -5.

Distance

The distance between a and b on the number line is |a-b|.

```
Distance between -3 and 5 = |-3 - 5| = |-8| = 8
```

13. Absolute Value

Definition: The absolute value of a real number is its distance from zero on the number line, so it is never negative.

```
|5| = 5; |-5| = 5; |0| = 0
```

Key Properties

```
|ab| = |a||b|

|a/b| = |a|/|b|, b != 0

|a+b| <= |a| + |b| (triangle inequality)
```

Common Use

Absolute value is used in distance, error, differences, constraints, and many programming problems.

14. Powers and Exponents

Definition: An exponent tells how many times a base is multiplied by itself. In a^n, a is the base and n is the exponent.

2^4 = 2 x 2 x 2 x 2 = 16

Exponent Laws

a^m x a^n = a^(m+n)

a^m / a^n = a^(m-n), a != 0

(a^m)^n = a^(mn)

(ab)^n = a^n b^n

(a/b)^n = a^n/b^n, b != 0

a^0 = 1, a != 0

a^(-n) = 1/a^n

Fractional Exponents

a^(1/n) = nth root of a

a^(m/n) = nth root of (a^m)

15. Roots and Radicals

Definition: The nth root of a number x is a number y such that y^n = x.

sqrt(25) = 5 because 5^2 = 25

cube root of 27 = 3 because 3^3 = 27

Simplifying Radicals

sqrt(50) = sqrt(25 x 2) = 5sqrt(2)

Basic Rules

sqrt(ab) = sqrt(a)sqrt(b) for non-negative a,b

sqrt(a/b) = sqrt(a)/sqrt(b), b > 0

Remember: Do not write sqrt(a+b) = sqrt(a)+sqrt(b). That rule is generally false.

16. Division Algorithm and Remainders

Definition: When an integer a is divided by a positive integer b, there exist unique integers q and r such that:

```
a = bq + r, where 0 <= r < b
```

Here a is the dividend, b is the divisor, q is the quotient, and r is the remainder.

Example: 29 divided by 6

29 = 6 x 4 + 5

Quotient = 4

Remainder = 5

Remainder in Programming

The modulo operator % returns the remainder for integer division in common programming languages.

```
29 % 6 = 5
```

17. Last Digit and Cyclicity

Definition: Cyclicity is the repeating pattern in the last digit (or other fixed suffix) of powers.

Last digit of 2^n

2^1 -> 2

2^2 -> 4

2^3 -> 8

2^4 -> 6

2^5 -> 2, and the pattern repeats.

Cycle length = 4.

For 2^100, compute 100 mod 4 = 0. Therefore use the fourth value in the cycle: 6.

Common Last-Digit Cycles

Base	Cycle
2	2,4,8,6
3	3,9,7,1
4	4,6
7	7,9,3,1
8	8,4,2,6
9	9,1

18. Modular Arithmetic

Definition: Modular arithmetic studies numbers according to their remainders after division by a positive modulus m.

> a mod m = r means a = mq + r, with 0 <= r < m

Congruence

We write a is congruent to b modulo m when m divides (a-b).

> a ≡ b (mod m) iff m | (a-b)

Example

17 ≡ 2 (mod 5), because 17 and 2 leave the same remainder when divided by 5.

Also, 17 - 2 = 15, and 5 divides 15.

Operations

> (a+b) mod m = [(a mod m)+(b mod m)] mod m
>
> (ab) mod m = [(a mod m)(b mod m)] mod m

These rules let you keep numbers small while computing very large expressions.

19. Decimal, Binary, Octal and Hexadecimal

Definition: A positional number system uses a base. The base tells how many digit symbols are available and how place values grow.

System	Base	Digits
Decimal	10	0-9
Binary	2	0-1
Octal	8	0-7
Hexadecimal	16	0-9 and A-F

Positional Representation

In base b: d_k b^k + d_(k-1)b^(k-1) + ... + d_1 b + d_0

Binary 1011

1011_2 = 1x2^3 + 0x2^2 + 1x2^1 + 1x2^0

= 8 + 0 + 2 + 1 = 11_10

Hexadecimal Digits

After 9, hexadecimal uses A=10, B=11, C=12, D=13, E=14, F=15.

20. Number System Conversions

Decimal to Binary

Repeatedly divide by 2 and record the remainders. Read the remainders from bottom to top.

13 to binary

13 / 2 = 6 remainder 1

6 / 2 = 3 remainder 0

3 / 2 = 1 remainder 1

1 / 2 = 0 remainder 1

Read upward: 1101_2.

Binary to Decimal

```
1101_2 = 1x8 + 1x4 + 0x2 + 1x1 = 13
```

Binary to Octal

Group binary digits into groups of 3 from the right.

```
101101_2 -> 101 101 -> 5 5 -> 55_8
```

Binary to Hexadecimal

Group binary digits into groups of 4 from the right.

```
10101111_2 -> 1010 1111 -> A F -> AF_16
```

Octal/Hexadecimal to Binary

Replace each octal digit with 3 bits and each hexadecimal digit with 4 bits.

21. Binary Arithmetic

Definition: Binary arithmetic performs arithmetic using only 0 and 1.

Binary Addition Rules

Operation	Result
0 + 0	0
0 + 1	1
1 + 0	1
1 + 1	10 (sum 0, carry 1)
1 + 1 + 1	11 (sum 1, carry 1)

Binary addition

1011

+ 0110

=10001

Binary Subtraction

Binary subtraction uses borrowing in the same way decimal subtraction does, but a borrowed 1 represents 2 in the current binary position.

Binary Multiplication

Binary multiplication is similar to decimal multiplication, but each partial product is either 0 or the original number.

22. Complements and Signed Representation

1's Complement

Definition: Flip every bit: 0 becomes 1 and 1 becomes 0.

```
101001 -> 010110
```

2's Complement

Definition: Take the 1's complement and add 1.

```
101001 -> 010110 -> 010111
```

2's complement is widely used to represent signed integers in computers.

Why 2's Complement?

- It gives one representation for zero in the standard fixed-width scheme.
- Addition and subtraction can use the same binary adder logic.
- Negative numbers can be represented naturally in fixed-width binary.

Range for n-bit Signed Integers

```
Range = -2^(n-1) to 2^(n-1)-1
```

For 8 bits, the range is -128 to 127.

23. Bitwise Operations

Definition: Bitwise operations manipulate the individual binary bits of integer values.

Operator	Meaning	Example
&	Bitwise AND	1010 & 1100 = 1000
|	Bitwise OR	1010 | 1100 = 1110
^	Bitwise XOR	1010 ^ 1100 = 0110
~	Bitwise NOT	~ flips every bit (width matters)
<<	Left shift	x << 1 shifts bits left
>>	Right shift	x >> 1 shifts bits right

Common DSA Uses

- Check odd/even: x & 1.
- Set, clear, toggle, and test individual bits.
- Represent subsets using bit masks.
- Use shifts in efficient integer calculations.
- XOR is useful in problems involving pairs, parity, and unique elements.

Remember: For signed integers, the exact behavior of right shift and NOT depends on the language and integer representation. Learn the language-specific rules before relying on them.

24. Euclidean Algorithm

Definition: The Euclidean algorithm computes the GCD of two integers using repeated remainders.

```
gcd(a,b) = gcd(b, a mod b)
```

gcd(252,105)

252 mod 105 = 42

105 mod 42 = 21

42 mod 21 = 0

Therefore gcd = 21.

Extended Euclidean Algorithm

The extended version also finds integers x and y satisfying ax + by = gcd(a,b). This is important for modular inverses and number theory.

```
ax + by = gcd(a,b)
```

25. Sieve of Eratosthenes

Definition: The Sieve of Eratosthenes is an algorithm for finding all prime numbers up to a limit n.

Procedure

- Create a list of integers from 2 to n.
- Start with p=2 and mark all multiples of p greater than p as composite.
- Move to the next unmarked number and repeat.
- Stop when p^2 > n.
- All unmarked numbers are prime.

Primes up to 20

Start with 2: mark 4,6,8,10,12,14,16,18,20.

Next 3: mark 6,9,12,15,18.

Next 5: mark 10,15,20.

Remaining numbers: 2,3,5,7,11,13,17,19.

> Time complexity: O(n log log n) for the standard sieve.

26. Fast Exponentiation

Definition: Fast exponentiation, also called binary exponentiation or exponentiation by squaring, computes a^n in O(log n) multiplications instead of O(n).

Core Idea

If n is even, a^n = (a^(n/2))^2. If n is odd, a^n = a x a^(n-1).

```
pow(a,2k) = pow(a,k)^2

pow(a,2k+1) = a x pow(a,k)^2
```

2^13

13 is odd. 2^13 = 2 x 2^12.

2^12 = (2^6)^2; 2^6 = (2^3)^2; 2^3 = 2 x 2^2.

The exponent is repeatedly halved, so the number of steps is logarithmic.

Modular Fast Power

In programming, reduce modulo m after each multiplication to keep values small.

```
result = (result x base) mod m
```

27. Prime Factorization in Programming

Definition: Prime factorization algorithms determine which prime numbers multiply together to form n.

Trial Division

Try dividing n by possible factors. After extracting a factor p, repeatedly divide it out.

A basic algorithm only needs to test p up to sqrt(n) because if n has a factor larger than sqrt(n), the matching factor is smaller than sqrt(n).

Example

84 -> divide by 2 twice -> 21 -> divide by 3 -> 7 -> prime

84 = 2^2 x 3 x 7

Why It Matters in DSA

- Count divisors.
- Compute divisor sums.
- Calculate Euler's totient.
- Solve factorization and number-theory problems.

28. Modular Inverse

Definition: The modular inverse of a modulo m is a number x such that ax is congruent to 1 modulo m.

$$a\,x \equiv 1 \pmod{m}$$

A modular inverse exists when gcd(a,m)=1.

Inverse of 3 modulo 7

Try x=5: 3x5=15.

15 mod 7 = 1.

Therefore 5 is the modular inverse of 3 modulo 7.

Extended Euclidean Method

If ax + my = 1, then x is an inverse of a modulo m. The extended Euclidean algorithm can find x.

Why Division is Different

Modulo division is not ordinary division. To divide by a modulo m, multiply by the modular inverse when the inverse exists.

$$a / b \pmod{m} \text{ is treated as } a \times \text{inverse}(b) \pmod{m}$$

29. Euler's Totient Function

Definition: Euler's totient function phi(n) counts the positive integers from 1 through n that are relatively prime to n.

phi(8)

Numbers from 1 to 8: 1,2,3,4,5,6,7,8.

Those co-prime to 8 are 1,3,5,7.

Therefore phi(8)=4.

Prime Formula

If p is prime, every number 1 through p-1 is co-prime to p.

```
phi(p) = p - 1
```

General Formula

If the distinct prime factors of n are p1, p2, ..., pk:

```
phi(n) = n x (1 - 1/p1) x (1 - 1/p2) x ... x (1 - 1/pk)
```

phi(12)

12 = 2^2 x 3.

phi(12) = 12(1-1/2)(1-1/3) = 4.

30. Fermat's Little Theorem

Definition: If p is prime and a is not divisible by p, then:

```
a^(p-1) ≡ 1 (mod p)
```

Equivalent Form

```
a^p ≡ a (mod p)
```

The first form is commonly used for reducing large powers modulo a prime.

Compute 2^10 mod 11

11 is prime.

By Fermat: 2^10 ≡ 1 (mod 11).

So the remainder is 1.

> Remember: Fermat's theorem has conditions. Do not apply the first form when p is not prime or when p divides a.

31. Chinese Remainder Theorem

Definition: The Chinese Remainder Theorem (CRT) gives conditions under which a system of simultaneous congruences has a unique solution modulo the product of pairwise coprime moduli.

Simple CRT system

$x \equiv 2 \pmod 3$

$x \equiv 3 \pmod 5$

Check numbers congruent to 2 mod 3: 2,5,8,11,14,...

14 is congruent to 4 mod 5, so continue; 8 is congruent to 3 mod 5.

Therefore $x \equiv 8 \pmod{15}$.

All solutions are $8 + 15k$ for integer k.

Where CRT Appears

- Number theory and modular arithmetic.
- Combining independent remainder constraints.
- Some algorithmic and cryptographic applications.

32. Useful Number Theory Identities

GCD and LCM

```
gcd(a,b) x lcm(a,b) = |ab|
```

Number of Divisors

If n = p1^a1 p2^a2 ... pk^ak, then the number of positive divisors is:

```
d(n) = (a1+1)(a2+1)...(ak+1)
```

Divisors of 60

60 = 2^2 x 3^1 x 5^1.

Number of divisors = (2+1)(1+1)(1+1) = 12.

Sum of Divisors

```
sigma(n) = [(p1^(a1+1)-1)/(p1-1)] x ... x [(pk^(ak+1)-1)/(pk-1)]
```

Parity Facts

- Even + even = even.
- Odd + odd = even.
- Even + odd = odd.
- Odd x odd = odd.
- Even x anything integer = even.

33. Common Mistakes

- 1. Treating 1 as a prime number. It is neither prime nor composite.
- 2. Forgetting that 2 is the only even prime.
- 3. Confusing factor and multiple. Factors divide; multiples are produced by multiplication.
- 4. Using the LCM instead of the GCD when the question asks for the greatest common divisor.
- 5. Forgetting that a fraction denominator cannot be zero.
- 6. Assuming every non-terminating decimal is irrational. Recurring decimals such as 1/3 are rational.
- 7. Applying sqrt(a+b)=sqrt(a)+sqrt(b). This is generally false.
- 8. Forgetting the condition gcd(a,m)=1 when finding a modular inverse.
- 9. Applying Fermat's Little Theorem without checking that the modulus is prime.
- 10. Mixing up binary grouping: 3 bits for octal and 4 bits for hexadecimal.
- 11. Ignoring integer width when reasoning about bitwise NOT and signed values.
- 12. Using floating-point arithmetic when an exact integer/modular answer is required.

34. Formula Sheet

Topic	Key Formula / Rule
Even	n=2k
Odd	n=2k+1
Division algorithm	a=bq+r, 0<=r<b
GCD	gcd(a,b)=gcd(b,a mod b)
GCD-LCM	gcd(a,b) x lcm(a,b)=|ab|
Exponents	a^m a^n=a^(m+n)
Exponents	a^m/a^n=a^(m-n)
Power of power	(a^m)^n=a^(mn)
Zero exponent	a^0=1 (a!=0)
Negative exponent	a^-n=1/a^n
Absolute value	|a| = distance from 0
Mod addition	(a+b) mod m = ((a mod m)+(b mod m)) mod m
Mod multiplication	ab mod m = ((a mod m)(b mod m)) mod m
Congruence	a ≡ b (mod m) iff m divides (a-b)
Signed n-bit range	-2^(n-1) to 2^(n-1)-1
Divisor count	d(n)=product(ai+1)
Totient prime	phi(p)=p-1
Fermat	a^(p-1) ≡ 1 (mod p), p prime and p does not divide a

35. Practice Set

Try these without looking at the answers. The goal is to build fluency, not just recognition.

1. Classify 0, -7, 3/4, sqrt(2), 11 as natural/whole/integer/rational/irrational/real where applicable.

2. Find all positive factors of 36.

3. Find the prime factorization of 360.

4. Find GCD(84,126).

5. Find LCM(12,18,30).

6. Simplify 3/4 + 5/8.

7. Convert 0.375 into a fraction.

8. State whether 7/40 has a terminating decimal expansion.

9. Simplify 2^5 x 2^3 / 2^4.

10. Simplify sqrt(72).

11. Find the remainder when 137 is divided by 12.

12. Find the last digit of 7^100.

13. Compute 12345 mod 9.

14. Convert 45 from decimal to binary.

15. Convert 110101_2 to decimal.

16. Convert 111011_2 to octal.

17. Convert 10111110_2 to hexadecimal.

18. Find the 2's complement of 00101101.

19. Evaluate 12 & 10 using binary.

20. Find gcd(252,105) using Euclid's algorithm.

21. List all primes <= 30 using the sieve.

22. Compute 3^100 mod 7 efficiently.

23. Find the modular inverse of 5 modulo 12.

24. Find phi(36).

25. Solve x ≡ 1 (mod 3), x ≡ 2 (mod 5).

Short Answer Key

1. 0 is whole/integer/rational/real; -7 is integer/rational/real; 3/4 is rational/real; sqrt(2) is irrational/real; 11 is natural/whole/integer/rational/real.

2. 1,2,3,4,6,9,12,18,36.

3. 360 = 2^3 x 3^2 x 5.

4. 42.

5. 180.

6. 11/8 = 1 3/8.

7. 3/8.

8. Yes.

9. 16.
10. 6sqrt(2).
11. 5.
12. 1.
13. 6.
14. 101101_2.
15. 53.
16. 73_8.
17. BE_16.
18. 11010011.
19. 8.
20. 21.
21. 2,3,5,7,11,13,17,19,23,29.
22. 4.
23. 5.
24. 12.
25. x ≡ 7 (mod 15).

Recommended Study Path for MCA + DSA

Use this chapter as a foundation rather than trying to finish the advanced number-theory section immediately.

Week / Phase	Focus	Goal
1	Types, factors, multiples, primes	Understand definitions and solve basic questions
2	Divisibility, GCD, LCM, fractions, decimals	Become comfortable with arithmetic reasoning
3	Powers, roots, remainders, cyclicity	Solve aptitude and DSA-style math problems
4	Binary, octal, hexadecimal, conversions	Build computer-science number-system knowledge
5	Complements, bitwise operations	Connect mathematics with programming
6	Euclid, sieve, fast power	Start algorithmic number theory
7+	Modular inverse, phi, Fermat, CRT	Study advanced DSA/competitive programming topics as needed

Remember: For interview preparation, prioritize definitions, GCD/LCM, divisibility, remainders, powers, binary conversions, 2's complement, bitwise operations, Euclidean algorithm, sieve, and fast exponentiation.

End of notes.